

ACTINIC KERATOSES

AT A GLANCE

Precancerous lesions on a spectrum from photodamaged skin to squamous cell carcinoma (SCC).

Strongest predictors of subsequent development not only of non-melanoma skin cancer but also melanoma.

Risk factors include individual susceptibility, cumulative ultraviolet (UV) radiation exposure, immunosuppression, prior history of skin cancers, and genetic syndromes.

Long-term and cumulative UV radiation exposure is the most important contributing factor for development of actinic keratoses (AKs).

Risk of progression of AK to SCC varies from less than 1 percent to 20 percent.

AKs should be treated because their course is unpredictable, they are often symptomatic, and they can progress to SCC if untreated.

Treatment methods include cryotherapy, topical agents, electrodesiccation and curettage, and photodynamic therapy.

Candidates for close longitudinal observation for prevention and early treatment.

Epidemiology

AKs are second only to acne vulgaris as the most common reason for patients to see a dermatologist. National Ambulatory Medical Care Survey data from 1990 to 1999 indicate that AKs were diagnosed in more than 47 million physician visits over that 10-year period and were found in 14 percent of patients visiting a dermatologist.

In the northern hemisphere, 11 percent to 25 percent of adults are estimated to have one or more AKs. In Australia, this prevalence rises to 40 percent to 60 percent of adults.

Risk Factors

The two major risk factors for AK development are individual susceptibility and cumulative UV radiation exposure.

Individual Susceptibility

- **Age:** Prevalence increases with age, ranging from less than 10 percent in white adults aged 20–29 years to up to 80 percent in those aged 60–69 years.
- **Gender:** Males have higher rates than females, likely due to greater cumulative sun exposure; this difference is more pronounced at younger ages.
- **Phenotype:** Fair skin that burns easily and freckles, blue or light-colored eyes, and red or blond hair are strongly associated with increased risk.
- **Immunosuppression:** Organ transplant recipients and others who are immunosuppressed have a significantly increased risk of developing AKs and SCC.
- **Genetic syndromes:** Conditions such as xeroderma pigmentosum, albinism, and possibly Bloom syndrome and Rothmund-Thomson syndrome predispose individuals to AKs due to impaired DNA repair or increased photosensitivity.

Environmental and Behavioral Risk Factors

- **Cumulative UV radiation exposure:** The most critical factor, including exposure from sunlight and tanning beds.
- **Prior history of AKs or skin cancers:** Individuals with previous lesions are at higher risk for developing new ones.

- **Geographic location:** Higher UV index and outdoor lifestyle (e.g., in Australia) contribute to increased prevalence.

TABLE 113-2 Risk Factors for Development of Actinic Keratoses

- **Individual susceptibility**
 - Older age
 - Male gender
 - Fair skin that easily burns and freckles
 - Blond or red hair
 - Light-colored eyes
- **Cumulative ultraviolet radiation exposure**
- **Immunosuppression**
- **Prior history of actinic keratoses or other skin cancers**
- **Genetic syndromes**
 - Xeroderma pigmentosum
 - Bloom syndrome
 - Rothmund-Thomson syndrome
 - Albinism